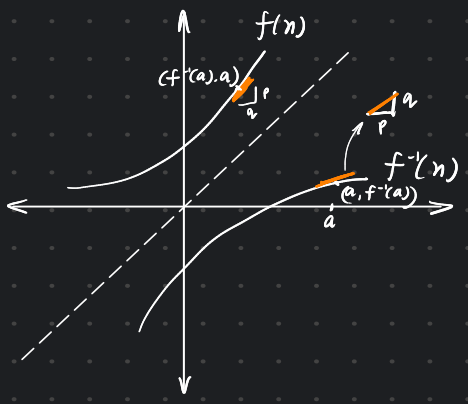
॥ श्रीगणेशाय नमः ॥

Derivatives of Inverse Functions



slopes will be reciprocals of one another

$$(f^{-1})'(a) = \frac{q}{p}$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$] Inverse derivative theorem

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Inverse function theorem

For functions of a single variable, the theorem states that if f is a continuously differentiable function with nonzero derivative at the point a; then f is injective (or bijective onto the image) in a neighborhood of a, the inverse is continuously differentiable near $b = f(a)$, and the derivative of the inverse function at b is the reciprocal of the derivative of f at a:

$$(f^{-1})'(b) = \frac{1}{f'(a)} = \frac{1}{f'(f^{-1}(b))}.$$